MARWAN EL SAABI GONZÁLEZ

DATA ANALYSIS & MACHINE LEARNING | MSC BIOINFORMATICS CANDIDATE

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PROFILE

Biologist and MSc Bioinformatics student (UDC, Feb 2027). I analyse data with Python, SQL and R, and have built personal projects in health data analysis, demand forecasting and machine learning. Seeking data analysis or data science internships or junior roles.

PROJECTS

Clinical Trials Analysis (SQL) | *SQL*, *DuckDB*, *Python*, *Streamlit*, *Plotly*

- SQL analysis of 600,000+ clinical trials (AACT, 40+ tables): ETL from PostgreSQL to DuckDB, reusable views and KPIs.
- Cleaned inconsistent records and classified 39,000 free-text stop reasons; 38% of failures were due to recruitment.

Retail Demand Forecasting | *Python*, *LightGBM*, *Pandas*, *Streamlit*

- Forecast weekly SKU demand 4 weeks ahead from 1M+ transactions using LightGBM with lag, price and calendar features.
- Beat naive, seasonal-naive and moving-average baselines by 4.8% wMAPE under rolling-origin backtesting.

Alzheimer's Drug Classifier (ML Virtual Screening) | *Python*, *Scikit-learn*, *XGBoost*, *RDKit*

- Curated 9.7k ChEMBL IC50 records into 4.3k unique molecules, encoded as 1024-bit Morgan fingerprints.
- Benchmarked RF, SVM, XGBoost and KNN with scaffold-split CV: 0.92 AUROC (a random split inflates it to 0.96).

Melanoma Detection (Classical Computer Vision) | *Python*, *OpenCV*, *Scikit-learn*, *Scikit-image*

- Automatic segmentation of dermoscopic lesions via Otsu, K-means and watershed, selected by IoU against ground truth.
- Extracted 40 shape, color and texture (LBP/GLCM) features; SVM reached 90.2% accuracy and 84% melanoma sensitivity.

PROFESSIONAL EXPERIENCE

- **Freelance AI Data Quality Analyst / TELUS Digital AI Data Solutions (remote)** | Sept 2026 to Present
- **Delivery Driver (Part-time) / Teigal Logística (Amazon Delivery Partner)** | June 2026 to Present
- **Production Operator / Stellantis España S.L, Vigo** | Oct 2024 to Oct 2025
- **Seasonal Farmworker (Grape Harvest) / Iman Temporing ETT, S.L, for Lagar de Fornelos** | Sept 2024

EDUCATION

University of A Coruña (Universidade da Coruña), A Coruña | Sept 2025 to Feb 2027
MSc in Bioinformatics Applied to Health Sciences
Master's thesis (in progress): information system on US packaged-food composition (UML, relational database, SQL, Power BI).
University of Vigo (Universidade de Vigo), Vigo | Jul 2024
BSc in Biology (Specialization: Biotechnology)

AREAS OF EXPERTISE

- **Programming & Data:** Python (NumPy, Pandas, OpenCV), R (Tidyverse, ggplot2, DESeq2), SQL, Excel, MATLAB, Bash.
- **Machine Learning:** Scikit-learn, XGBoost, LightGBM, Neural Networks, Cross-Validation, Forecasting, LLM APIs (Gemini).
- **Tools & Infrastructure:** Linux, Git/GitHub, Docker, Conda, Nextflow, High-Performance Computing (SLURM).

LANGUAGES

Spanish and Galician (native), English (B2, CertAcles)

PUBLICATION

Alarcón-Ríos L., **Saabi González M.**, Velo-Antón G. (2026). Melanism in salamanders: effects of insularity, sex and size on dorsal colouration in *Salamandra salamandra*. *Journal of Zoology*. doi:10.1111/jzo.70133